
SpatialR: A Benchmark Evaluating LLMs’ Spatial Reasoning Skills

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Abstract

1 Large language models (LLMs) have been prompted to do any and all types
2 of reasoning tasks for users; spatial reasoning being one of them, before truly
3 knowing their limitations. Spatial reasoning can be defined as the ability to mentally
4 visualize, manipulate, and rotate 2D and 3D objects, understand how parts fit
5 together, and their relationship in space. Spatial reasoning can be used in various
6 settings and has various topics. Many benchmarks exist to measure one specific
7 area of spatial reasoning skills, but not many encompasses the broad topic of
8 spatial reasoning completely. To effectively evaluate these models’ true spatial
9 reasoning abilities, we have built a benchmark, SpatialR, that incorporates topics
10 related to both two-dimensional and three-dimensional spatial reasoning. This
11 benchmark was developed as a broader, more comprehensive benchmark that
12 covers both 2D and 3D reasoning tasks. We generated these tasks using templates
13 we created of our own task ideas with word banks with groupings of keywords.
14 These templates allowed us to quickly create multiple tasks that assess the same
15 spatial reasoning ability. Our benchmark was implemented using the Benchtools,
16 a tool that aids researchers in setting up benchmarks. Using this benchmark, we
17 evaluated several small models, including Google’s Gemma 4 family, Qwen 3
18 and 3.6, and mistral, and discovered that these models have significant difficulties
19 applying even the simplest transformations to given examples. Our results from this
20 benchmark’s evaluation show that current LLMs still lack robust spatial reasoning
21 despite general language ability. Code, dataset, and benchmark can be found on
22 our Github repository: <https://github.com/evalAI2eval/SpatialReasoning.git>

23 1 Introduction

24 Spatial reasoning refers to the ability to understand how objects exist and change within space. It
25 involves recognizing an object’s position, orientation, and movement, as well as understanding how
26 those properties change when the object is moved or viewed from a different perspective. People
27 rely on this type of reasoning frequently in everyday situations. It appears when navigating through
28 a building, interpreting diagrams, or imagining how an object would look after being turned or
29 repositioned. These skills are especially useful in STEM-related occupations as well as every day
30 life. Monjoree and Yan ([n. d.]) Many puzzles and visual problems depend on the same type of
31 thinking, where the difficulty comes from mentally tracking how an object changes while maintaining
32 an accurate understanding of its orientation and position.

33 Despite the rapid progress of multi-modal language models, spatial reasoning remains an area where
34 performance can still be inconsistent. Models often perform well when answers can be inferred
35 from patterns in training data, but may struggle when the task requires maintaining a consistent
36 understanding of where an object is as it moves through space. This is because of their behavior to
37 detect and imitate language patterns instead of building reasoning on any type of ground proof. Madala
38 et al. ([n. d.]) Williams and Huckle (2024) LLMs are known for their strong linguistic capabilities but
39 their spacial reasoning abilities remain limited due to their learning being confined to the text-based
40 representations collected from their language-only training. Liu et al. (2025) Small variations in
41 wording or orientation can sometimes lead to different answers even when the reasoning process
42 should remain the same. This possibly means that models may depend on pattern recognition more
43 than making a representation of the spatial relationship of the object and its surroundings.

44 Many benchmarks focus on the ability to understand language related to spatial topics rather than the
45 ability to perform spatial tasks. Bisk et al. (2017) Rodionov et al. (2026) Anand and Shareghi (2025)
46 Each of the previously-published benchmark focus on one or a few specific spatial topics. While
47 these benchmarks provided us with valuable insights on the LLMs ability to complete one type of
48 tasks, it is difficult to use already existing benchmarks to make general conclusions about LLMs’
49 spatial awareness and understanding

50 We introduce a benchmark designed to evaluate spatial reasoning in both text-only and multi-modal
51 settings. The goal is to measure whether a model can maintain a consistent representation of an
52 object as it is moved in space. The tasks are structured around sequences of spatial operations where
53 an object begins in a default position and then goes through one or more transformations. These
54 problems resemble the type of reasoning used when solving spatial puzzles, where the challenge lies
55 in tracking the transformation rather than identifying the object itself.

56 The benchmark will include a variety of tasks that test the same spatial reasoning ability throughout
57 different formats. Some questions will be presented entirely through text descriptions, while others
58 will include simple images that represent an object and its orientation. Each task will require the
59 model to follow a sequence of spatial changes and determine the final orientation or relative position
60 of the object. By constraining the answer space and varying the transformations used, the benchmark
61 aims to isolate spatial reasoning ability rather than allowing models to rely on memorized patterns.
62 The tasks focus on common spatial transformations such as rotations, directional movement, and
63 relative positioning, requiring the model to maintain a consistent internal representation of an object’s
64 orientation and location across multiple steps.

65 2 Related Work

66 Several benchmarks have attempted to measure LLMs’ capabilities with respect to spatial awareness,
67 but they primarily focus on spatial intelligence and perception. One of the earliest work in spatial
68 reasoning datasets was the GroundedLanguage dataset Bisk et al. (2017) published in 2017. Although
69 it predates the rise of large language models, this benchmark included tasks that encompasses the
70 skills previous benchmarks evaluated, as well as introducing higher-level concepts such as mirroring
71 and balancing. While models can often identify objects within a given image, they struggle to
72 understand these objects in a three-dimensional space. For example, a model could identify a car in
73 an image, but it might struggle to understand its position if we ask it to consider the transformation
74 that would occur if it was rotated.

Due to the fact that spatial awareness is an extremely broad concept, every benchmark measuring this concept has its own unique definition. These benchmarks must adopt a definition that narrows their objective enough to make it both achievable and novel. SpatialViz-Bench Wang et al. (2026) achieves this using a framework of four key sub-abilities: mental rotation, mental folding, visual penetration, and mental animation. The majority of tasks in this benchmark involve abstract shapes rather than functional objects. The role of functional objects in spatial manipulation tasks remains underexplored. This benchmark evaluates performances on a variety of spatial awareness exercises, but there are recent benchmarks researchers developed for a specific type of spatial exercises. SpatialBench Xu et al. (2025) is another benchmark that attempts to build a collection of tasks to build a broader definition of spatial-awareness. This benchmark contains 15 question types each mapping to one of their five defined cognitive levels; Observational, Topology and Relation, Symbolic Reasoning, Causality, and Planning. PlanQARodionov et al. (2026) is an example of this as it evaluates LLM’s spatial reasoning using 2D floor plans represented in structured text-based formats. This benchmark include questions that require assessments of visibility, distance computation, evaluation of placement feasibility, and reasoning about spatial constraints.

There are several benchmarks that contain tasks similar to our benchmark’s, but many do not have a variety of exercises as large as SpatialR. CubeBenchAnand and Shareghi (2025) is a benchmark that can be used to evaluate LLMs’ models ability to solve a Rubrik’s cube. Because solving a Rubrik’s cube is a complex task, CubeBench’s tasks are broken down into several sub-tasks such as cube face recognition. Cube face recognition determines if the LLM can identify the directional faces (front, left, etc.) from an image of a Rubrix’s cube laid out in a net diagram.

Our benchmark shifts the focus toward active spatial reasoning. Specifically, it evaluates a model’s ability to mentally manipulate objects, predict the outcomes of spatial transformations, and understand how those changes impact real-world interactions. By comparing our approach to prior benchmarks, we highlight both the strengths of existing models and the gaps our benchmark is designed to address. Our benchmark provides a more rigorous test of spatial reasoning and artificial common sense.

3 Methodology / Approach

3.1 Preliminaries and Backgrounds

This benchmark contains several exercises to measure LLM’s ability to use common-sense spatial reasoning. We define common-sense reasoning as the ability to correctly infer relative orientation, rotation, and spatial displacement of objects under geometric transformations, using either textual descriptions alone or text grounded in visual input. This includes; mental rotation, relative positioning, directional movement inference, and transformation composition. As of now, no visual or multi modal inputs exist, but multimodal functionality will be the focus of future work.

- **Mental transformation** is the ability to infer an object orientation after rotation, sequential composition of transformation. This includes both single-step transformations (e.g., rotate 90° clockwise) and sequential composition of transformations (e.g., rotate 90° clockwise and then 180°). Correctly completing these types of exercises requires maintaining an internal representation of orientation and correctly updating it under rule-based operations.
- **Relative positioning** is the ability to define an object’s spatial location with respect to another object or reference point. For example, in the statement “The ball is to the left of the chair.”, the model must correctly interpret directional relationships within a shared coordinate frame. This includes reasoning about object-centered versus global reference frames and maintaining consistency when the reference object changes.
- **Directional movement** is the ability to interpret and apply movement commands, such as up, forward, clockwise, or counterclockwise. These exercises require mapping keywords to geometric operations and updating positional states accordingly. Movement reasoning may involve translating a point across a grid, following the direction of an arrow, or applying rotation to a facing object.
- **Transformation composition** is the ability to apply multiple spatial transformations sequentially to one object and identify its final state. This tests whether the models are performing true compositional reasoning or relying on shallow pattern recognition. For

example, combining rotation and translation requires maintaining intermediate state representations and applying transformation rules in order.

Including these four categories of capabilities will evaluate whether a model maintains internally consistent spatial representations that are stable across textual and visual modalities. Our exercises for this benchmark will be split into two types of exercises; text-only spatial reasoning exercises and image + text multi modal exercises. Each exercise will have a standardized input-output structure, meaning that all exercises will follow a uniform prompt template. Each exercise will explicitly instruct the model to select a single-word answer from a predefined answer bank. This benchmark will test models performance on these topics using various exercises. We will construct it as a controlled benchmark composed of discrete exercises across multi-modal settings. Each exercise will evaluate a specific skill under the large umbrella of spatial reasoning. By using this benchmark, we will evaluate how effectively large language models understand and manipulate physical space layouts, and object relationships.

3.2 Benchmark Structure

This benchmark was designed to include two main components: generic transformations and algorithmic transformations. The exercises for the generic transformations are intended to measure the spatial awareness of objects through single step and multi step transformations, also including questions about the functionality of objects after being transformed. The algorithmic transformation exercises serve as a way to measure and evaluate the LLM’s spatial capacities and competence in a more challenging, algorithmic domain. The benchmark is performed 30 times, with scores averaged across all 30 iterations being presented for the final results.

3.2.1 Generic Transformations

A series of templated spatial exercises are presented to the model, with an answer bank containing valid answers. Models are prompted to respond with a single word from the answer bank. Scoring is binary, questions given the correct answer from the given answer bank are correct. Incorrect answers, or answers containing words from outside of the answer bank are marked incorrect. Scoring is displayed as total correct/incorrect, as well as by tags (2-Dimensional, 3-Dimensional, transformation only, "common sense" style question, specific object transformed). These questions will be either multi modal (image + text) or text-based, and fit four categories:

1. General Spatial Transformations
 - (a) Translations (2D and 3D)
 - (b) Rotations (2D and 3D)
 - (c) Combined Translations and Rotations (3D)
 - (d) Post-transformation Object Functionality (3D), referred to as "*common sense*"
2. Algorithmic Transformations

3.2.2 Algorithmic Transformations

Spatial exercises that involve reasoning about a series of algorithmic transformations to objects are also presented to the model. There are two types of multi-step transformation exercises:

1. 2x2 Rubik’s Cube Puzzles
2. Dice Rotations

with the intent of evaluating forward transformation exercises and inverse transformation exercises. Forward transformation exercises give the model a text-based representation of the initial state of an object and a series of transformations to be performed on the object. Pierella et al. (2019) For these exercises, the model will be prompted to respond with a text-based representation of the resulting state of the object after the transformations. Scoring is binary, only given answers that exactly match the correct answer are correct. Scoring will be displayed as total correct/incorrect. Inverse transformation exercises provide the model with a text-based representation of the initial

177 state of an object and a goal state which can be reached by performing transformations on the
 178 object. Pierella et al. (2019) For these exercises, the model will be prompted to respond with the series
 179 of transformations required to reach that goal state. Options for the transformations that may be used
 180 are provided in an answer bank and may be repeated multiple times in the output list.

181 3.2.3 Exercise Generation

182 To allow for a wide variety of exercises and address potential sampling bias by hand authoring each
 183 transformation, this benchmark was constructed pseudo-randomly (seed based) using text based
 184 templates and generative scripts. This allowed for simple and parameterized inputs to vary the
 185 difficulty, modality, and dimensionality of the exercises to examine how each factor affects the
 186 LLMs’ ability to produce correct answers. Due to time constraints, detailed experiments across these
 187 dimensions were not conducted, but functionality remains for future work to build off of. Additionally,
 188 these generative scripts were designed to remove as much ambiguity as possible, only generating
 189 exercises that have clear and singular answers from the provided answer bank. An example of this

Figure 1 shows an example of a template-generated exercise used in the benchmark.

<u>Simple Transformation (2d)</u>	<u>Rotate Die (3d)</u>
Answer the following question with a one word response using the valid response pool: If an {arrow} facing {right} is rotated {90 degrees} {clockwise}, which direction is it now facing? Valid Answers: up, down, left, right	Given the following directional faces of a numerical die and the values on those directional faces {Front: 1, Back: 6, Up: 4, Down: 3, Left: 2, Right: 5}, what are the values on the directional faces after the following transformations? {rotate front face 90 degrees counterclockwise} Respond in the format that was used to produce the initial state of the die.

Figure 1: Example of a template-generated benchmark exercise. The model is given a standardized prompt with a fixed answer bank and must return the correct final orientation after transformation.

190

191 3.3 Dataset

192 This benchmark contains 200 generated exercises from 10 templates, and 30 hand-crafted exercises
 193 from 3 templates (the *common sense* variants.) To our knowledge, these exercises contain little
 194 to no reference to open-source materials or other available materials. We believe that creating
 195 these exercises ourselves rather than importing them from sources allows these exercises to be
 196 unfamiliar to the models. This allows the benchmark to be evaluated based on its ability to complete
 197 spatial reasoning exercises rather than identifying exercises it has already been exposed to during
 198 training. Conceptually, these exercises can be split into evaluating orientation/relative position
 199 through rotations and transformations, as well as evaluating object functionality after performing said
 200 transformations. These categories are further broken down by dimension, quantity of transformations,
 201 and transformation type.

202 3.3.1 Rotations and Translations

203 This exercise category contains 5 templates of exercises to evaluate basic two-dimensional and
 204 three-dimensional spatial reasoning through singular or multiple consecutive rotations, translations,
 205 or combination of both performed on objects about specified axis or through given vectors, given
 206 starting orientations. The difference between two-dimensional and three-dimensional exercises is
 207 only the inclusion of a z axis as an option for transformation, not necessarily the dimensionality of
 208 the objects being rotated.

If a desktop monitor is turned sideways and moved 4 feet to the side of a workstation, can it still be viewed from the original seating position?
 <Answer: No>

Figure 2: A sample common sense translation task

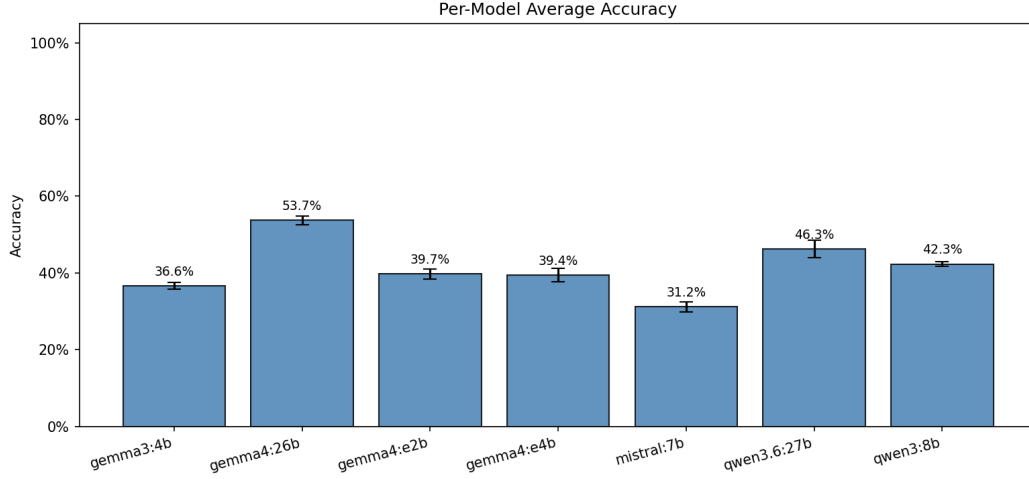


Figure 3: Average model accuracy across 30 benchmarks for the Generic Transformations category

3.3.2 Common Sense exercises

This exercise category mirrors the Rotations/Translations category with the inclusion of a *common sense* style question on the functionality of a given object after transformations have been applied. An example can be seen in Figure 2

4 Experiments and Results

The benchmark was run 30 times for each model, with results averaged across all 30 iterations to account for potential variation in performance due to the stochastic nature of LLMs.

4.1 Generic Transformations

Average model performance across the 30 performed benchmarks was as expected, with higher parameter models out performing smaller models as seen in Figure 3. Prompt adherence for this category was not an issue, and performance among each individual benchmark repeat was largely consistent across all models.

Despite the expectation of performance degrading when going from 2D to 3D, and when going from single transformation to multi transformation exercises, some smaller models performed better on both jumps. Looking at Table 2, we see that gemma3:4b actually increases in accuracy when going from single three dimensional transformation exercises to multi step three dimensional transformation exercises. This also was the case for 2d rotations as shown in Table 1, but not for 2d translation tasks.

This is likely due to an unintended consequence of data aggregation and tagging, as we aggregate results based on tag, so our categorical/tag-based analysis includes *common sense* results in their other more broad parent categories (i.e. common sense questions after 2d multi rotations also contain the tag '2d multi rotation', shared with pure 2d multi rotation tasks).

Model	Total Avg. Acc.	2D Rot Single	2D Rot Multi	2D Trans. Single	2D Trans. Multi
gemma3:4b	0.3662	0.1850	0.3050	0.6000	0.4350
gemma4:26b	0.5371	0.7233	0.2917	0.6250	0.4650
gemma4:e2b	0.3971	0.3500	0.2017	0.5500	0.4917
gemma4:e4b	0.3943	0.4233	0.2017	0.4267	0.4017
mistral:7b	0.3117	0.1217	0.1483	0.5683	0.4117
qwen3.6:27b	0.4626	0.5117	0.2283	0.5350	0.4767
qwen3:8b	0.4229	0.2183	0.2133	0.6000	0.5633

Table 1: Two-dimensional results for single and multi step rotation/translation transformations, with the best performing accuracy in bold. Total Avg. Acc. represents accuracy across all categories to allow for baseline comparison.

Model	Total Avg. Acc.	3D Rot. Single	3D Rot. Multi	3D Trans. Single	3D Trans. Multi
gemma3:4b	0.3662	0.1017	0.1717	0.3100	0.4150
gemma4:26b	0.5371	0.5267	0.3133	0.5700	0.2267
gemma4:e2b	0.3971	0.6000	0.1233	0.2750	0.3100
gemma4:e4b	0.3943	0.5500	0.1617	0.3367	0.4867
mistral:7b	0.3117	0.2767	0.1467	0.3100	0.2167
qwen3.6:27b	0.4626	0.6817	0.1850	0.3250	0.2683
qwen3:8b	0.4229	0.5567	0.0550	0.3450	0.4717

Table 2: Three-dimensional results for single and multi step rotation/translation transformations, with the best performing accuracy in bold. Total Avg. Acc. represents accuracy across all categories to allow for baseline comparison.

4.1.1 Common Sense Transformations

Another interesting result is the performance on the *common sense* style questions. As shown in Table 4, *common sense* exercise performance generally exceeded that of all other categories, with gemma4:26b consistently achieving perfect accuracy in the three dimensional combined category. These exercises did contain fewer examples, were hand crafted, and potentially meaningfully differed in structure from the other generated examples, so further exploration is required.

4.2 Algorithmic Transformations

Unfortunately, these tasks spawned continuous, 'broken' chain of thought style responses from larger models, preventing completion of this category. Due to time constraints, further work needs to be done focusing on prompt adherence, exercise formulation, and response token limitations.

5 Limitations, Conclusion, and Future Work

5.1 Limitations

There are a great many of limitations in this benchmark spanning its concept, design, and implementation. To non-comprehensively highlight some of the key limitations that we see as valuable to address first:

1. Our transformation exercises do not directly translate to human spatial awareness
2. Our tagging and analysis approach needs more granularity to improve analysis quality
3. Phrasing between categories requires standardization/refinement to ensure our own verbiage does not cause performance difference between categories
4. Our templates/transformations are simple and procedural, and the benchmark could easily be completed with high accuracy if a model were to treat the exercises as math problems using tool calls
5. We lack any sort of human baseline
6. We lack multi modality for an inherently multi modal task

Model	Total Avg. Acc.	2D Combination	3D Combination
gemma3:4b	0.3662	0.5067	0.3250
gemma4:26b	0.5371	0.8550	0.5283
gemma4:e2b	0.3971	0.3817	0.3183
gemma4:e4b	0.3943	0.3150	0.2717
mistral:7b	0.3117	0.2400	0.2017
qwen3.6:27b	0.4626	0.6083	0.4300
qwen3:8b	0.4229	0.4067	0.3467

Table 3: Two-dimensional and three-dimensional combined transformation exercise results, with the best performing accuracy in bold. Total Avg. Acc. represents accuracy across all categories to allow for baseline comparison.

Model	Total Avg. Acc.	3D Rot. CS	3D Trans. CS	3D Comb. CS
gemma3:4b	0.3662	0.5133	0.8000	0.4000
gemma4:26b	0.5371	0.6033	0.5000	1.0000
gemma4:e2b	0.3971	0.4767	0.6500	0.8033
gemma4:e4b	0.3943	0.4433	0.6300	0.8467
mistral:7b	0.3117	0.6300	0.6933	0.5633
qwen3.6:27b	0.4626	0.7100	0.6233	0.8067
qwen3:8b	0.4229	0.7000	0.7000	0.7733

Table 4: Two-dimensional and three-dimensional *common sense* exercise results, with the best performing accuracy in bold. Total Avg. Acc. represents accuracy across all categories to allow for baseline comparison.

5.2 Conclusion

Using SpatialR, we see that many models struggle with spatial reasoning tasks but are able to consistently answer others. Most of the models tested performed in line with expectations when examined comparatively; higher parameter counts are generally correlated with higher overall performance when looking at models from the same generations. Results that may appear interesting do appear to be an unintended consequence of our tagging approach, which will be addressed in future work. From overall performance we hypothesize that these tested models do not maintain stable internal representations of spatial states across various spatial transformations.

5.3 Future Work

There are many possibilities for the expansion of this benchmark. Our set of exercises contains tasks that only assess one spatial reasoning skill at a time. A possible future direction for this work is creating tasks that include a combination of spatial reasoning skills, as well as more abstract skills, including various modalities. Additionally, homogenizing our current set of exercises between categories is critical to ensuring our own varying approach does not produce unintended differences in performance. It would also be valuable to benchmark more state of the art (SOTA) models, and allow for chain of thought in the benchmark itself. Preliminary and small scale tests that the team has performed seem to indicate that SOTA models may perform significantly better on our exercises.

References

- Dhruv Anand and Ehsan Shareghi. 2025. Cube Bench: A Benchmark for Spatial Visual Reasoning in MLLMs. doi:10.48550/arXiv.2512.20595 arXiv:2512.20595 [cs].
- Yonatan Bisk, Kevin J. Shih, Yejin Choi, and Daniel Marcu. 2017. Learning Interpretable Spatial Operations in a Rich 3D Blocks World. doi:10.48550/arXiv.1712.03463 arXiv:1712.03463 [cs].
- Weichen Liu, Qiyao Xue, Haoming Wang, Xiangyu Yin, Boyuan Yang, and Wei Gao. 2025. Spatial Reasoning in Multimodal Large Language Models: A Survey of Tasks, Benchmarks and Methods. doi:10.48550/arXiv.2511.15722 arXiv:2511.15722 [cs].
- Siddarth Madala, Risham Sidhu, and Tianjiao Yu. [n. d.]. Spatial Awareness in LLMs. ([n. d.]).

- 281 Uttamasha Monjoree and Wei Yan. [n. d.]. AI’s Spatial Intelligence: Evaluating AI’s Understanding
282 of Spatial Transformations in PSVT:R and Augmented Reality. ([n. d.]).
- 283 Camilla Pierella, Maura Casadio, Ferdinando A. Mussa-Ivaldi, and Sara A. Solla. 2019. The dynamics
284 of motor learning through the formation of internal models. *PLoS Computational Biology* 15, 12
285 (Dec. 2019), e1007118. doi:10.1371/journal.pcbi.1007118
- 286 Fedor Rodionov, Abdelrahman Eldesokey, Michael Birsak, John Femiani, Bernard Ghanem, and
287 Peter Wonka. 2026. FloorplanQA: A Benchmark for Spatial Reasoning in LLMs using Structured
288 Representations. doi:10.48550/arXiv.2507.07644 arXiv:2507.07644 [cs].
- 289 Siting Wang, Minnan Pei, Luoyang Sun, Cheng Deng, Yuchen Li, Kun Shao, Zheng Tian, Haifeng
290 Zhang, and Jun Wang. 2026. SpatialViz-Bench: A Cognitively-Grounded Benchmark for Diag-
291 nosing Spatial Visualization in MLLMs. doi:10.48550/arXiv.2507.07610 arXiv:2507.07610
292 [cs].
- 293 Sean Williams and James Huckle. 2024. Easy Problems That LLMs Get Wrong. doi:10.48550/
294 arXiv.2405.19616 arXiv:2405.19616 [cs].
- 295 Peiran Xu, Sudong Wang, Yao Zhu, Jianing Li, Gege Qi, and Yunjian Zhang. 2025. SpatialBench:
296 Benchmarking Multimodal Large Language Models for Spatial Cognition. doi:10.48550/ARXIV.
297 2511.21471 Version Number: 3.